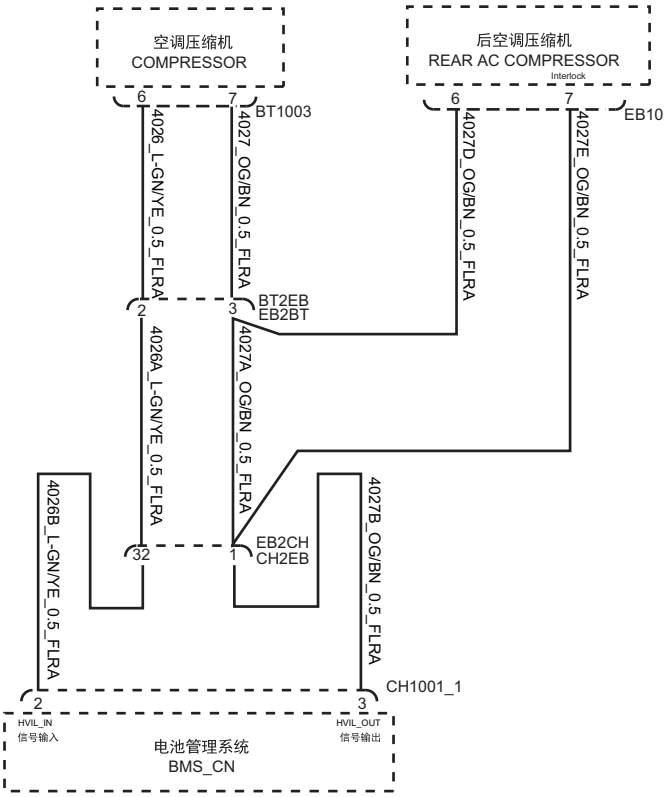
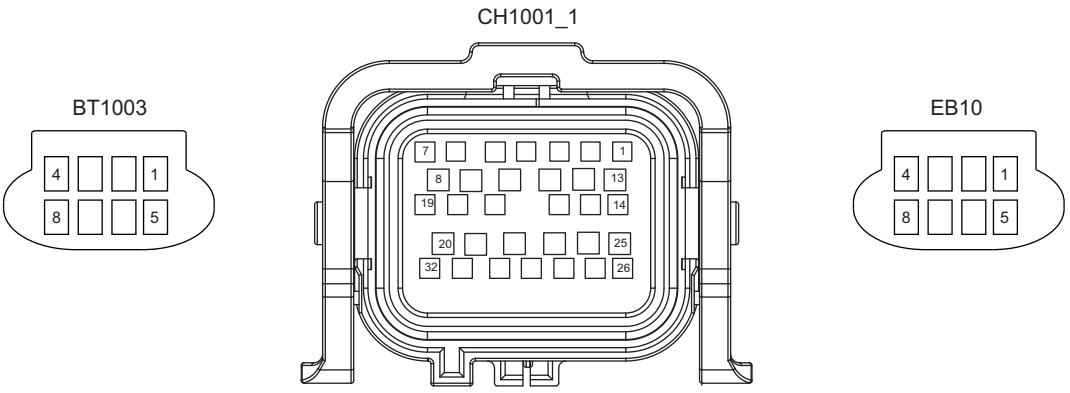
P0A0A-00、 P0A0B-00、 P0A0C-00、 P0A0D-00

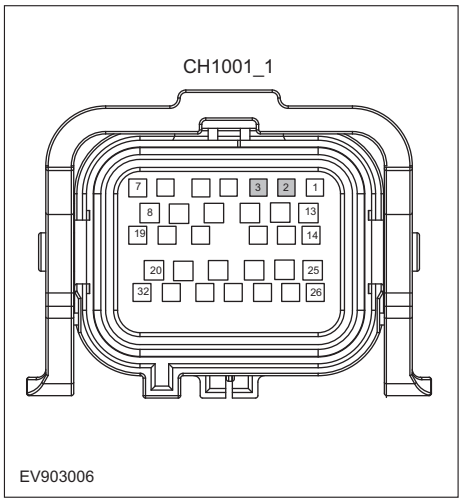
Fault diagnosis code
P0A0A-00：High voltage interlock loop 1 open circuit
P0A0B-00：High voltage interlock loop invalid
P0A0C-00：High voltage interlock loop 1 short to GND
P0A0D-00：High voltage interlock loop 1 short to source power
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BMS fault • Interlock instrument fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0A0A-00
High voltage interlock loop 1 open, loop resistance value is greater than the set range
P0A0B-00
High voltage interlock loop 1 short to source power
P0A0C-00
High voltage interlock loop 1 short to GND
P0A0D-00
If the state of HV interlock is greater than 7 times in the 13 check periods, it is a HV interlock loop error, but it is not a “HV interlock loop short”or “HV interlock loop short”or “HV interlock loop short to ground”
To set the effect of a fault code condition
High voltage component does not work properly
Description of circuit diagram
BMS always monitors high voltage interlock circuit. Meanwhile,it will check the connector for virtual connection, short circuit, open circuit and the connection rationality with other signals. When fault occurs, the BMS module will save the fault code corresponding to that fault and enable safety mode.

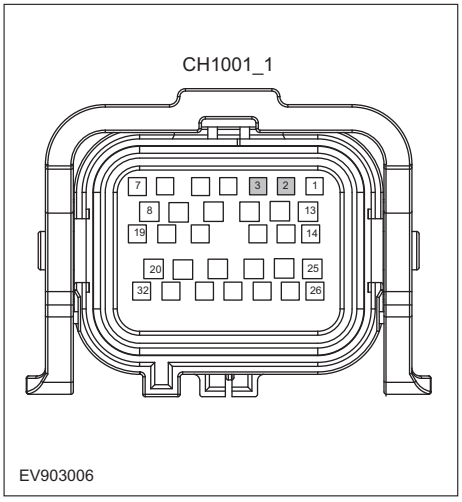
Circuit diagram

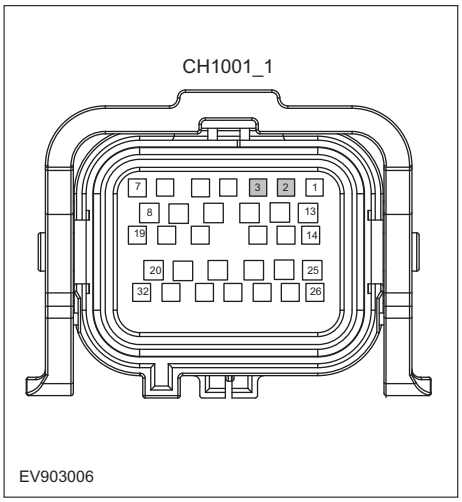




EV903005

Troubleshooting steps	
Step 1	General inspection.
	1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of BMS for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BMS with diagnostic apparatus. 4 Is there any other fault code except for P0A0A-00、P0A0B-00、P0A0C-00、P0A0D-00?
	YES → Refer to:DTC Summary list(BMS).
NO ↓	
Step 3	Check BMS interlock circuit. (Open circuit)
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug CH1000_1 of BMS. 3 Measure the resistance between the harness terminals of BMS. Measuring circuit: resistance between terminal 2 and terminal 3 on plug CH1000_1. **Standard value: < 1Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 4	Check BMS interlock instrument. (Module terminal)
	1 Measure the resistance between the harness terminals of BMS(Module terminal). Measuring circuit: resistance between terminal 2 and terminal 3 on plug CH1000_1. **Standard value: < 1Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check BMS interlock circuit. (Short circuit)
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness terminal of BMS and ground. Measuring circuit: resistance between terminal 2 on plug CH1000_1 and ground. Measuring circuit: resistance between terminal 3 on plug CH1000_1 and ground. **Standard value: ∞** 3 Measure the voltage between the harness terminal of BMS and ground. Measuring circuit: voltage between terminal 2 on plug CH1000_1 and ground. Measuring circuit: voltage between terminal 3 on plug CH1000_1 and ground. **Standard value: ≈ 0V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Test whether BMS is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug CH1000_1 of BMS. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BMS plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BMS.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	